

Lab: Computational Psychiatry and Clinical Applications

Learning Goal: Apply Active Inference to analyze psychiatric conditions as specific patterns of inference failure, and propose treatment targets.

Part 1: Diagnostic Mapping

Exercise: For each condition, specify the computational failure, the neural substrate, the key symptom it explains, and the treatment target:

Condition	Computational Failure	Key Neural Substrate	Symptom Explained	Treatment Target
Schizophrenia (positive)	↑ sensory PE precision (aberrant salience)	Mesolimbic DA pathway, superior temporal gyrus	Auditory hallucinations	Reduce DA-mediated error precision
Schizophrenia (negative)	↓ motivational precision / ↓ policy evaluation	Mesocortical DA pathway, PFC	Avolition, flat affect	Restore policy precision (motivation)
Major depression	Pessimistic priors + ↓ positive PE precision	vmPFC, anterior insula, raphe nuclei	Persistent sadness, anhedonia	Update priors via behavioral activation
Generalized anxiety	↑ expected environmental volatility	Locus coeruleus, amygdala, ACC	Chronic worry, hypervigilance	Reduce expected volatility (exposure, SSRIs)
Panic disorder	Catastrophic interoceptive inference	Anterior insula, PAG	Panic attacks	Recalibrate interoceptive model
Autism (sensory)	Inflexibly ↑ sensory PE precision	Primary sensory cortices	Sensory overwhelm, detail focus	Environmental accommodations, flexible precision training
Autism (social)		STS, mPFC, TPJ		

Condition	Computational Failure	Key Neural Substrate	Symptom Explained	Treatment Target
	↓ social prediction precision		Social communication difficulties	Explicit rule teaching, social scripts
Addiction	Hijacked policy precision (drug-seeking dominates)	Ventral striatum, OFC	Compulsive drug use despite harm	Restore policy competition (contingency management)
OCD	↑ precision on error signals + inability to resolve	Cortico-striatal loops, ACC	Intrusive thoughts, compulsive rituals	ERP therapy (tolerate unresolved PEs)
PTSD	Traumatic model locked by terror priors	Amygdala, hippocampus, vmPFC	Flashbacks, hyperarousal	Reconsolidation-based exposure, EMDR

Now choose one condition and write a 300-word analysis tracing the full computational pathway from neural mechanism to symptom experience.

Write your response here

Part 2: Case Formulation

Learning Goal: Develop an Active Inference case formulation for a clinical scenario.

Exercise: Read the following case and formulate it in Active Inference terms:

Case: A 32-year-old woman presents with 6 months of increasing anxiety. She avoids social situations, constantly checks her phone for reassurance texts from friends, and has developed insomnia. She reports feeling "like something terrible is about to happen" most of the time, and experiences her heart racing at work. She had a stable childhood but reports a recent breakup as the trigger.

1. **Generative model analysis:** What are her current priors? (Threat = likely; social rejection = likely; bodily symptoms = dangerous)
2. **Precision profile:** Which prediction errors have excessive precision? (Interoceptive errors, social threat errors)
3. **Policy analysis:** What policies is she using to minimize free energy? (Avoidance = reduce exposure to threatening sensory input; reassurance-seeking = reduce uncertainty)
4. **Maintenance mechanisms:** How do these policies maintain the problem? (Avoidance prevents model updating; reassurance provides temporary PE reduction but maintains the prior that threat is real)
5. **Treatment targets:** What would need to change in her generative model?

Write a complete Active Inference case formulation (400 words).

Write your response here

Part 3: Treatment Design

Learning Goal: Design treatment based on computational psychiatry principles.

Exercise: For two of the following conditions, design a treatment protocol that targets the specific computational failure:

Option A — Depression: Design a behavioral activation protocol. Explain: (a) why behavioral activation generates positive prediction errors, (b) how these errors update pessimistic priors, (c) what role therapist guidance plays in precision weighting, (d) how to handle the patient's tendency to down-weight positive evidence.

Option B — Social anxiety: Design an exposure protocol. Explain: (a) what the maladaptive prior is ("others will judge me negatively"), (b) how social exposure generates corrective prediction errors, (c) why avoidance maintains the problem, (d) how precision affects whether exposure is effective.

Option C — PTSD: Design a reconsolidation-based intervention. Explain: (a) how the traumatic memory is an overly precise prior, (b) how reactivation makes the memory labile, (c) how introducing novel information during the reconsolidation window updates the traumatic model, (d) what makes this potentially more effective than standard exposure.

Write your response here

Part 4: Pharmacological Analysis

Learning Goal: Analyze psychopharmacology through Active Inference.

Exercise: For each medication class, explain the computational mechanism:

Medication	Primary Action	Active Inference Interpretation
Antipsychotics (haloperidol)	Block D2 receptors in striatum	Reduce precision of sensory PEs → less aberrant salience, fewer hallucinations/delusions
SSRIs (fluoxetine)	Increase serotonin availability	Modulate temporal precision → reduce urgency of aversive signals, increase patience/long-horizon thinking
Benzodiazepines (diazepam)	Enhance GABA inhibition	Globally reduce precision of prediction errors → reduced anxiety but also reduced sensitivity
Stimulants (methylphenidate)	Increase DA/NE availability	Increase policy precision → improved sustained attention and task focus
Psychedelics (psilocybin)	5-HT2A agonism, disrupt DMN	Relax overly rigid priors → enable model revision for depression, PTSD, addiction
Ketamine	NMDA antagonism	Acutely disrupt top-down predictions → enable new learning patterns; rapid antidepressant effect

Write your response here

Part 5: Reflection

In 300 words, reflect: Computational psychiatry promises personalized treatment based on individual computational profiles. But psychiatric diagnosis is currently based on symptom checklists (DSM-5), not computational measurements. What would need to change in clinical practice for computational psychiatry to become practical? What barriers exist?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Diagnostic mapping	Condition → computational failure
2	Case formulation	Individual clinical analysis
3	Treatment design	Computationally targeted intervention
4	Pharmacological reasoning	Medication mechanisms
5	Clinical reflection	Implementation barriers